Taken together, these perspectives suggest that users may respond more favorably to refusals that exhibit topical alignment and conversational relevance.

> ***H3a: Refusals that exhibit higher conversational alignment with the user's prompt are associated with comparably higher satisfaction ratings.***

A second feature that may shape user perception is response length. While concise answers can enhance clarity, overly short refusals—particularly those lacking reasoning or elaboration—may be perceived as low-effort or unengaged. Longer refusals, by contrast, might include justifications, context, or constructive alternatives, which could signal greater communicative effort and care.

This idea aligns with the effort heuristic (Kruger et al., 2004), which suggests that people tend to interpret more elaborate responses as more thoughtful or trustworthy, even when the content is similar. Similarly, Information Richness Theory (Daft & Lengel, 1986) posits that richer messages are especially valued in ambiguous or sensitive communication contexts—conditions that often accompany refusal scenarios. Recent work in LLM evaluation has also shown that both human and model-based raters demonstrate a length bias, consistently favoring longer responses in pairwise comparisons (Huang et al, 2024; Gu et al. 2024), possibly because verbosity is interpreted as a proxy for thoroughness or helpfulness.

These insights suggest that longer refusals may be received more positively, particularly when they provide reasoning, or maintain conversational engagement.

> ***H3b: Refusals with longer response length are associated with comparably higher satisfaction ratings.***

2.4. Topic Sensitivity and Ethical Refusals

Beyond how refusals are phrased or presented, another important factor influencing user satisfaction may be what the refusal is about—particularly in the case of ethical refusals. While alignment with safety guidelines is crucial, user reactions can vary depending on the sensitivity of the underlying topic. Refusals on high-risk prompts—such as those involving illegal activity, hate speech, or self-harm—may be interpreted as justified and responsible, whereas refusals on less ethically charged prompts—such as fictional roleplay, political satire, or edgy humor—may feel excessive, unjustified, or intrusive.

This distinction is supported by research on trust in AI systems, which highlights that users are more accepting of constraints when they perceive them as legitimate, proportional, and motivated by benevolent intent. For instance, Kieslich et al. (2021) find that users tend to trust AI systems that clearly align with ethical principles and safety norms. Similarly, Binns et al. (2018) and Gillespie (2018) emphasize that perceptions of fairness and transparency play a key role in user acceptance—particularly when restrictions affect expressive or exploratory content.

Users may tolerate refusals that block dangerous or illegal requests, but react negatively when moderation targets prompts that seem playful, ambiguous, or socially permissible. In these cases, refusals can be experienced not only as unjustified but as unsettling—especially when they appear vague or moralizing. This aligns with findings from Eslami et al. (2019), who describe how opaque or unexplained restrictions can trigger discomfort, especially when users feel surveilled or misjudged by the system.

Recent LLM alignment benchmarks such as SORRY-Bench (Xie et al., 2024) and MultiTaskBench (Jan et al., 2025) reflect the difficulty of fine-tuning refusals across content types. While refusals are essential for enforcing safety, overgeneralized filters risk undermining user trust—particularly when users do not perceive their prompts as sensitive in the first place.

> ***H4: Ethical refusals on less sensitive topics (e.g., humor, fiction, political satire) are more negatively associated with user satisfaction than refusals on more sensitive topics (e.g., illegal activities or hate speech).***

3. Methodology

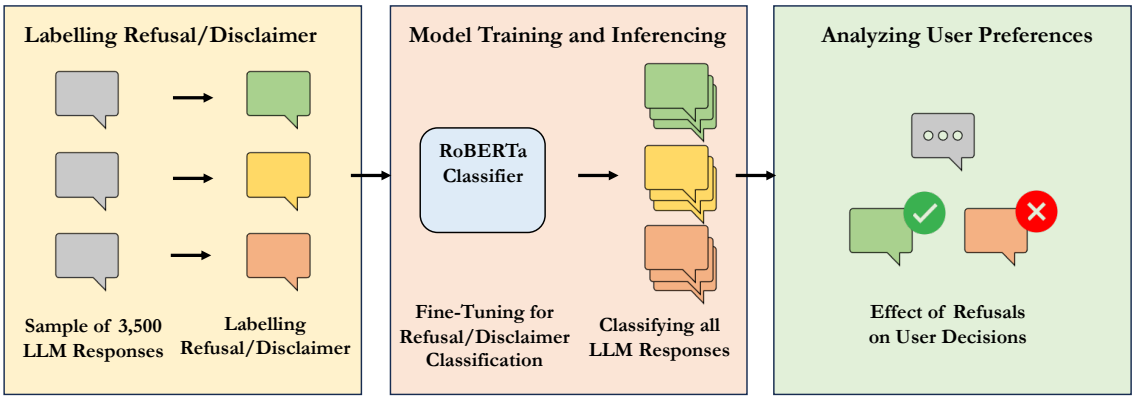

Figure 1. Overview of Main Methodological Pipeline

Figure 1 illustrates the main methodological pipeline of this study. We begin by manually annotating a sample of 3,500 LLM responses with a taxonomy of refusal and disclaimer behaviors. These annotations are used to fine-tune a RoBERTa-Large classifier, which is then applied to classify all responses in the Chatbot Arena dataset. This enables large-scale identification of refusal types across nearly 50,000 model comparisons. Finally, we analyze how these response types relate to user preferences by examining win/loss decisions in pairwise model comparisons.

In addition to this main pipeline, we enrich the analysis with two complementary components: (i) semantic features of refusal phrasing, including response length and textual similarity; and (ii) prompt sensitivity, as measured by the OpenAI Moderation API.

3.1. Data

To investigate the impact of content moderation by LLMs on user satisfaction, we leverage data from Chatbot Arena, a widely used benchmarking platform for conversational AI models (Chiang et al., 2024). The dataset consists of 57,477 conversation pairs, with each pair comprising responses from two distinct models to a shared user prompt. Users are tasked with selecting their preferred response or declaring a tie if neither model is favored. This user feedback serves as the indicator of user satisfaction.

A significant majority (approximately 86%) of all interactions in the dataset are single-turn conversations, meaning they consist of a single user prompt followed by two model responses. These one-turn interactions offer a clear, isolated view of user preferences, minimizing the potential influence of follow-up clarifications or context from the user. To maintain the interpretability of user decisions, we exclude multi-turn conversations (those involving multiple user-model interactions) from our analysis. This choice ensures that user preferences are directly linked to the initial model responses rather than follow-up exchanges, which could introduce complexities like jailbreaking attempts or context refinement. After applying this filtering criterion, our final dataset consists of 49,938 conversation pairs, each with a single user prompt and two corresponding model responses.

3.2. Labeling Response Refusals and Disclaimers

To the best of our knowledge, no existing classifier distinguishes between the different types of refusals exhibited by large language models (LLMs). While benchmarks such as SORRY-Bench (Xie et al., 2024) offer valuable tools for evaluating how LLMs handle ethically sensitive prompts, they focus exclusively on scripted, high-risk questions designed to trigger refusal. This makes them effective for stress-testing alignment, but less suited for analyzing how refusals emerge in organic, user-generated contexts. In real-world settings, refusals can occur across a broad range of queries—and it becomes essential not only to detect when models refuse, but also to disentangle whether those refusals stem from ethical concerns or from functional limitations.

To address this gap, we developed a taxonomy of refusal behavior based on a two-dimensional classification scheme: the first dimension identifies the motivation behind the boundary (ethical vs. technical), and the second distinguishes whether the model fully rejects the task (refusal) or continues to engage while noting constraints (disclaimer).

The first axis reflects the reason why an LLM refuses (or qualifies) its response to a prompt. Drawing on prior work in human–AI interaction and LLM safety research, we distinguish between two basic categories:

- **Ethical refusals and disclaimers** arise when the model declines a request or qualifies its response due to concerns about safety, legality, or appropriateness. These boundaries are widely discussed in the context of AI alignment and content moderation, where LLMs are expected to avoid generating harmful or socially unacceptable outputs (Kieslich et al., 2021; Weidinger et al., 2021; Xie et al., 2024). Typical examples include refusals to provide instructions for illegal activities or advice on self-harm.